Reasoning under Uncertainty

Probabilistic Inference and Causal Reasoning

CS3263

Foundations of Artificial Intelligence

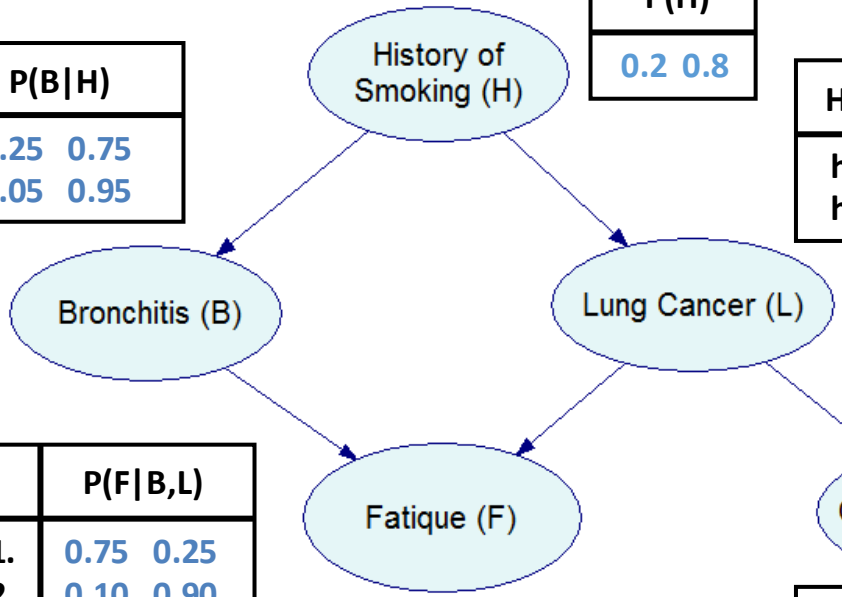
Version 4.0

Topics

- Bayesian network inference
  - Enumeration
  - Variable elimination
- Causal reasoning
  - Bayesian causal networks
  - Do-Calculus

Bayesian Network Inference

Review



P(H)
0.2 0.8

H	P(B|H)
h1	0.25 0.75
h2	0.05 0.95

H	P(L|H)
h1	0.00300 0.99700
h2	0.00005 0.99995

B L	P(F|B,L)
b1 l1.	0.75 0.25
b1 l2.	0.10 0.90
b2 l1	0.50 0.50
b2 l2.	0.05 0.95

L	P(C|L)
l1	0.60 0.40
l2	0.02 0.98

Inference in Bayesian Networks

- Inference is just calculation of conditional distributions!
- Given the values of a set of evidence variables **E** = **e**, find:
- Prior or posterior distribution for set of query variables
  - $P(\boldsymbol{X}|\boldsymbol{e})$: e.g., **e = (smokes, bronchitis), X = lung cancer**
  - e.g., Diagnosis, prediction, inter-causal reasoning, etc.
- Most probable values of set of query variables
  - Maximum A Posteriori (MAP)/Most Probable Explanation (MPE)
  - $\text{argmax}_{\boldsymbol{Y}} P(\boldsymbol{Y}|\boldsymbol{e})$
  - e.g., Classification, disease combination, etc.

Joint Probability

- Answering the above questions requires the joint probability distribution
- What can you compute with the joint distribution $P(X_1, X_2, \dots, X_n)$?
  - Joint probability: $P(X_1, X_2)$
  - Marginal Probability: $P(X_1)$, $P(X_2)$, $P(X_3)$, $P(X_4)$, $P(X_1, X_2)$, $P(X_1, X_3)$, etc.
  - Conditional distributions: $P(X_1 | X_2, X_3, X_4)$ (Predictions)
  - Most likely explanations: $\max_{x_1, x_3} P(X_1 = x_1, X_3 = x_3 \,| X_2, X_4)$

Example

- Given the joint probability table of:
  - History of smoking (H)
  - Bronchitis (B)
  - Lung cancer (L)
  - Fatigue (F)
  - Chest X-ray (C)
- What are the conditional probabilities of the person 1) having bronchitis and 2) having lung cancer:
  - $P(b_1|h_1, f_1, c_1)$?
  - $P(l_1|h_1, f_1, c_1)$?
- What is the most likely disease combination?
  - $\text{argmax}_{\{b,l\}} P(b, l|\ h_1, f_1, c_1)$?

H	B	L	F	C	P(HBLFC)
h1	b1	l1	f1	c1	0.00006750
h1	b1	l1	f1	c2	0.00004500
h1	b1	l1	f2	c1	0.00002250
h1	b1	l1	f2	c2	0.00001500
h1	b1	l2	f1	c1	0.00009970
h1	b1	l2	f1	c2	0.00488530
h1	b1	l2	f2	c1	0.00089730
h1	b1	L2	f2	c2	0.04396770
h1	b2	L1	f1	c1	0.00013500
h1	b2	l1	f1	c2	0.00009000
h1	b2	l1	f2	c1	0.00013500
h1	b2	l1	f2	c2	0.00009000
h1	b2	l2	f1	c1	0.00014955
h1	b2	l2	f1	c2	0.00732795
h1	b2	l2	f2	c1	0.00284145
h1	b2	l2	f2	c2	0.13923105
h2	b1	l1	f1	c1	0.00000090
h2	b1	l1	f1	c2	0.00000060
h2	b1	l1	f2	c1	0.00000030
h2	b1	l1	f2	c2	0.00000020
h2	b1	l2	f1	c1	0.00008000
h2	b1	l2	f1	c2	0.00391980
h2	b1	l2	f2	c1	0.00071996
h2	b1	l2	f2	c2	0.03527824
h2	b2	l1	f1	c1	0.00001140
h2	b2	l1	f1	c2	0.00000760
h2	b2	l1	f2	c1	0.00001140
h2	b2	l1	f2	c2	0.00000760
h2	b2	l2	f1	c1	0.00075996
h2	b2	l2	f1	c2	0.03723814
h2	b2	l2	f2	c1	0.01443928
h2	b2	l2	f2	c2	0.70752462

Example

H	B	L	F	C	P(HBLFC)
h1	b1	l1	f1	c1	0.00006750
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h1	b1	l2	f2	c1	0.00089730
h1	b1	L2	f2	c2	0.04396770
h1	b2	L1	f1	c1	0.00013500
h1	b2	l1	f1	c2	0.00009000
h1	b2	l1	f2	c1	0.00013500
h1	b2	l1	f2	c2	0.00009000
h1	b2	l2	f1	c1	0.00014955
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$$P(b_1|h_1,c_1,f_1) = \frac{P(b_1,h_1,c_1,f_1)}{P(h_1,c_1,f_1)}$$

$$= \frac{\sum_l P(h_1,b_1,l,f_1,c_1)}{\sum_b \sum_l P(h_1,b,l,f_1,c_1)}$$

$$= \frac{\sum \text{blue rows}}{\sum \text{blue \& orange rows}}$$

$$= 0.370116$$

Example

H	B	L	F	C	P(HBLFC)
h1	b1	l1	f1	c1	0.00006750
h1	b1	l1	f1	c2	0.00004500
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$$P(b|h_1, c_1, f_1) = \frac{P(b, h_1, c_1, f_1)}{P(h_1, c_1, f_1)} = \alpha \times P(b, h_1, c_1, f_1)$$

$$P(b_1|h_1, c_1, f_1) = \alpha \times P(b_1, h_1, c_1, f_1) = \alpha \times \sum \text{blue rows}$$
$$= \alpha \times 0.0001672$$

$$P(b_2|h_1, c_1, f_1) = \alpha \times P(b_2, h_1, c_1, f_1) = \alpha \times \sum \text{orange rows}$$
$$= \alpha \times 0.00028455$$

Since $P(b_1|h_1, c_1, f_1) + P(b_2|h_1, c_1, f_1) = 1$, we can normalize them:

$$P(b_1|h_1, c_1, f_1) = 0.0001672/(0.0001672 + 0.00028455)$$
$$P(b_2|h_1, c_1, f_1) = 0.00028455/(0.0001672 + 0.00028455)$$

Inference by Enumeration

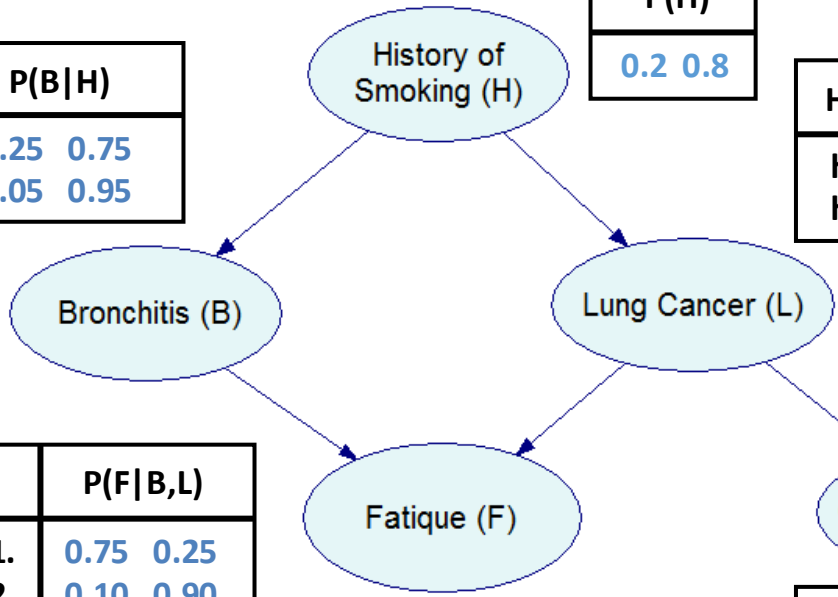
- Any conditional probability can be computed by summing terms from the full joint distributions
- In general, a query P(X|**a**) can be answered by marginalization using the equation:

$$P(X \mid \boldsymbol{a}) = \frac{P(X, \boldsymbol{a})}{P(\boldsymbol{a})} = \alpha P(X, \boldsymbol{a}) = \alpha \sum_{\boldsymbol{y}} P(X, \boldsymbol{a}, \boldsymbol{y})$$

Implications

- Why is this a good idea?
  - Any marginal and conditional probability can be read off a joint distribution table by summing over the relevant rows
- Why is this a bad idea?
  - No. of entries in the joint distribution table can be huge!
  - 32 in this example; imagine a problem with 100 variables!
- In general:
  - Want to calculate the conditional probabilities more efficiently, using much less storage for the joint distribution "table"!

Recall: Example



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h2	0.05	0.95

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h1	0.00300	0.99700
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B	L	P(F|B,L)	
b1	l1.	0.75	0.25
b1	l2.	0.10	0.90
b2	l1.	0.50	0.50
b2	l2.	0.05	0.95

L	P(C|L)	
l1	0.60	0.40
l2	0.02	0.98

$$P(B,C,F,H,L) = P(H)P(B \mid H)P(L \mid H)P(F \mid B,L)P(C \mid L)$$

Joint Probability Distribution

H	B	L	F	C	P(HBLFC)
h1	b1	l1	f1	c1	0.00006750
h1	b1	l1	f1	c2	0.00004500
h1	b1	l1	f2	c1	0.00002250
h1	b1	l1	f2	c2	0.00001500
h1	b1	l2	f1	c1	0.00009970
h1	b1	l2	f1	c2	0.00488530
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h1	b1	l2	f2	c2	0.04396770
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h2	b1	l1	f1	c2	0.00000060
h2	b1	l1	f2	c1	0.00000030
h2	b1	l1	f2	c2	0.00000020
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h2	b1	l2	f2	c2	0.03527824
h2	b2	l1	f1	c1	0.00001140
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h2	b2	l2	f2	c1	0.01443928
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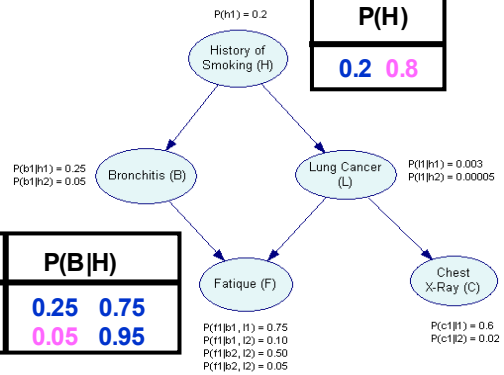

P(H)	
0.2	0.8

H	P(L|H)	
h1	0.00300	0.99700
h2	0.00005	0.99995

H	P(B|H)	
h1	0.25	0.75
h2	0.05	0.95

B	L	P(F|B,L)	
b1	l1	0.75	0.25
b1	l2	0.10	0.90
b2	l1	0.50	0.50
b2	l2	0.05	0.95

L	P(C|L)	
l1	0.60	0.40
l2	0.02	0.98

$$P(h_2,b_1,l_2,f_2,c_2) = P(h_2)P(b_1|h_2)P(l_2|h_2)P(f_2|b_1,l_2)P(c_2|l_2)$$

$$= 0.8 * 0.05 * 0.99995 * 0.90 * 0.98$$

$$= 0.03527825$$

Efficient Inference

H	B	L	F	C	P(HBLFC)
h1	b1	l1	f1	c1	0.00006750
h1	b1	l1	f1	c2	0.00004500
h1	b1	l1	f2	c1	0.00002250
h1	b1	l1	f2	c2	0.00001500
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h1	b2	l2	f2	c1	0.00284145
h1	b2	l2	f2	c2	0.13923105
h2	b1	l1	f1	c1	0.00000090
h2	b1	l1	f1	c2	0.00000060
h2	b1	l1	f2	c1	0.00000030
h2	b1	l1	f2	c2	0.00000020
h2	b1	l2	f1	c1	0.00008000
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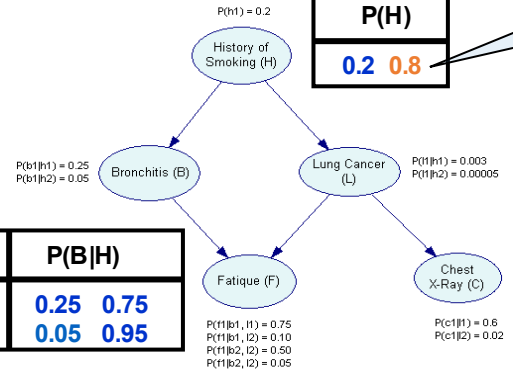

P(H)	
0.2	0.8

$P(h_2)$
Nice and easy

H	P(L|H)	
h1	0.00300	0.99700
h2	0.00005	0.99995

H	P(B|H)	
h1	0.25	0.75
h2	0.05	0.95

B	L	P(F|B,L)	
b1	l1	0.75	0.25
b1	l2	0.10	0.90
b2	l1	0.50	0.50
b2	l2	0.05	0.95

L	P(C|L)	
l1	0.60	0.40
l2	0.02	0.98

$$P(h_2) = \underset{b}{\square}\ \underset{l}{\square}\ \underset{f}{\square}\ \underset{c}{\square}\ P(h_2, b, l, f, c)$$

Efficient inference

H	B	L	F	C	P(HBLFC)
h1	b1	l1	f1	c1	0.00006750
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h2	b2	l2	f2	c1	0.01443928
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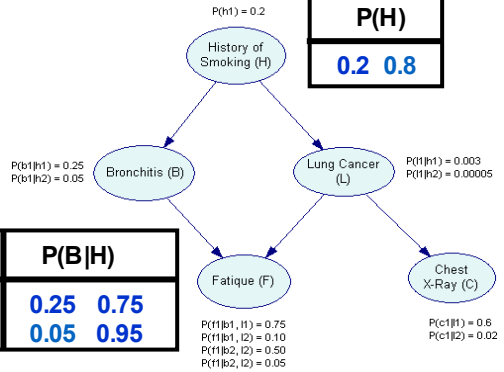

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0.2	0.8

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h1	0.00300	0.99700
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h1	0.25	0.75
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B	L	P(F|B,L)	
b1	l1	0.75	0.25
b1	l2	0.10	0.90
b2	l1	0.50	0.50
b2	l2	0.05	0.95

L	P(C|L)	
l1	0.60	0.40
l2	0.02	0.98

But how about $P(f_2)$?

When Does BN Help in Efficient Inference?

- Let k be maximum number of dependencies of any node in a BN. Then, the number of parameters is bounded by:

$$n \times 2^k$$

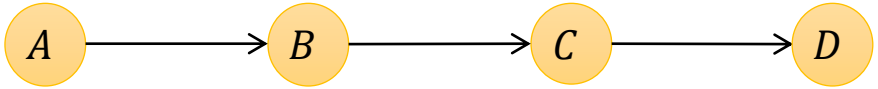
- For $n = 100$ variables and a BN without any arcs, i.e., $k = 0$
  - Joint probability distribution table would be huge.
  - But all the marginals P(X) would be readily available
  - Also easy to calculate any combination of things.
- For $n = 100$ variables and a BN with all the possible arcs, i.e., $k = 99$
  - There is one node s without children in such a network
  - The conditional probability table for P(s|99 other variables) has 2^{99} rows.
  - Little help for inference in this case.

Variable Elimination

An illustrative example

Factor algebra

An illustrative example



$P(A)$	
a_1	a_2
0.7	0.3

	$P(B\|A)$	
	b_1	b_2
a_1	0.6	0.4
a_2	0.3	0.7

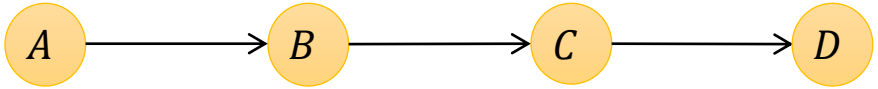
	$P(C\|B)$	
	c_1	c_2
b_1	0.3	0.7
b_2	0.1	0.9

	$P(D\|C)$	
	d_1	d_2
c_1	0.8	0.2
c_2	0.9	0.1

What is the marginal probability of D, i.e., $P(D = d)$?

$$P(d) = \sum_a \sum_b \sum_c P(a, b, c, d) = \sum_a \sum_b \sum_c P(d|c)P(c|b)P(b|a)P(a)$$

An illustrative example



	a	b	c
1	a_1	b_1	c_1
2	a_1	b_1	c_2
3	a_1	b_2	c_1
4	a_1	b_2	c_2
5	a_2	b_1	c_1
6	a_2	b_1	c_2
7	a_2	b_2	c_1
8	a_2	b_2	c_2

A naïve approach: list down all possible cases of (a, b, c):

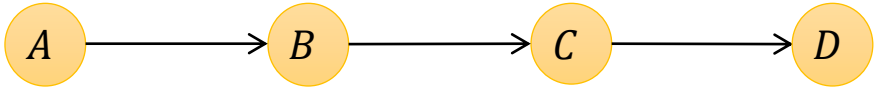
$$P(d) = \sum_a \sum_b \sum_c P(d|c)P(c|b)P(b|a)P(a)$$

Complexity: 2^3

If we have n variables to marginalize out:

$$2^n$$

An illustrative example



$$P(d) = \sum_a \sum_b \sum_c P(a,b,c,d) = \sum_a \sum_b \sum_c P(d|c)P(c|b)P(b|a)P(a)$$

$$= \underbrace{\sum_c P(d|c) \underbrace{\sum_b P(c|b) \underbrace{\sum_a P(a)P(b|a)}_{\text{Step 1}}}_{\text{Step 2}}}_{\text{Step 3}}$$

An illustrative example

$$A \rightarrow B \rightarrow C \rightarrow D$$

$P(A)$	
a_1	a_2
0.7	0.3

	$P(B\|A)$	
	b_1	b_2
a_1	0.6	0.4
a_2	0.3	0.7

	$P(a)P(b\|a)$	
	b_1	b_2
a_1	0.6 x 0.7	0.4 x 0.7
a_2	0.3 x 0.3	0.7 x 0.3

Sum over a

$\phi_1(b)$	
b_1	b_2
0.51	0.49

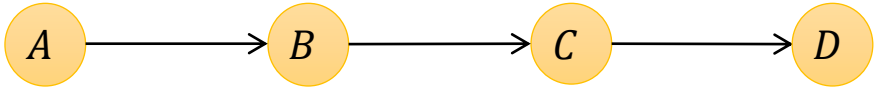
Step 1

$$\phi_1(b) = \sum_a P(a)P(b|a)$$

Complexity: 2^2

General complexity: 2^k

An illustrative example



$$P(d) = \sum_c P(d|c) \sum_b P(c|b) \sum_a P(a)P(b|a)$$

$$= \underbrace{\sum_c P(d|c) \underbrace{\sum_b P(c|b)\, \phi_1(b)}_{\text{Step 2}}}_{\text{Step 3}}$$

An illustrative example

$$A \rightarrow B \rightarrow C \rightarrow D$$

$\boldsymbol{\phi_1(b)}$	
b_1	b_2
0.51	0.49

	$\boldsymbol{P(C\|B)}$	
	c_1	c_2
b_1	0.3	0.7
b_2	0.1	0.9

	$\boldsymbol{P(c\|b)\phi_1(b)}$	
	c_1	c_2
b_1	0.3 x 0.51	0.7 x 0.51
b_2	0.1 x 0.49	0.9 x 0.49

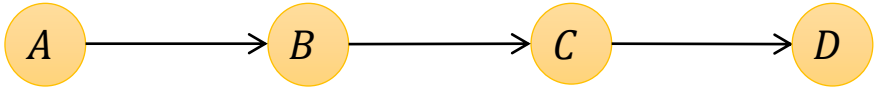
Step 2

$$\phi_2(c) = \sum_b P(c|b)\,\phi_1(b)$$

$\boldsymbol{\phi_2(c)}$	
c_1	c_2
0.202	0.798

Complexity: 2^k

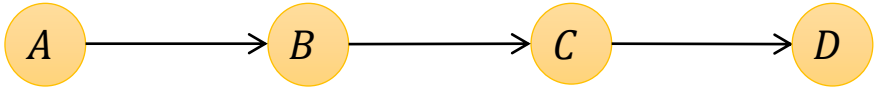
An illustrative example



$$P(d) = \sum_c P(d|c) \sum_b P(c|b) \sum_a P(a)P(b|a)$$

$$= \underbrace{\sum_c P(d|c)\,\phi_2(c)}_{\text{Step 3}}$$

An illustrative example



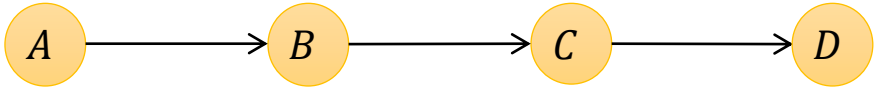
$\boldsymbol{\phi_2(c)}$	
c_1	c_2
0.202	0.798

	$\boldsymbol{P(D\|C)}$	
	d_1	d_2
c_1	0.8	0.2
c_2	0.9	0.1

Step 3 The calculations are similar to those of step 1 and 2.

$$P(d) = \sum_{c} P(d|c)\,\phi_2(c)$$

An illustrative example



Since there are 3 steps, the total complexity: $2^k + 2^k + 2^k = 3 \times 2^k$

When there are n variables to marginalize out, the total complexity: $n \times 2^k$

Inference by enumeration: 2^n
Variable elimination:
$n \times 2^k$

Variable elimination is only helpful when $k \ll n$

Variable Elimination

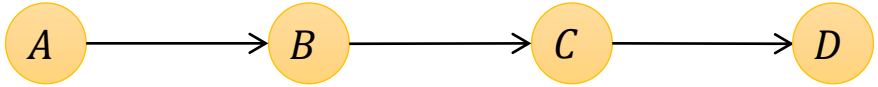
- Main ideas:
  - Re-arranging the sums and products that need to be computed
  - Calculate once and save results for later use
  - Improve enumeration by eliminating repeated calculations
  - A form of dynamic programming – with memoization!
- The variable elimination algorithm repeatedly performs two factor operations:
  - Multiplication and Marginalization

Factor (or Potential) Algebra

- Define factor (potential) ϕ to be a (non-negative) real-function over a domain of variables X:
  - $\phi: \Omega_X \rightarrow \Re^+$
- The domain or scope of a factor ϕ is denoted by $\mathrm{dom}(\phi)$
  - $\phi(X_1, \dots, X_n)$ indicates that $X_1, \dots, X_n$ are in $\mathrm{dom}(\phi)$
- A probability distribution is a special factor over a set of variables X:
  - $P(X)$ is a factor whose elements $\sum_x P(x) = 1$
  - $P(X|Y)$ is a factor whose elements $\sum_x P(x|y) = 1$
  - Domain of factor $P(A, B|C)$ is $dom\big(P(A, B|C)\big) = \{A, B, C\}$
- Factors can be viewed as matrices and implemented as tables
- Factors can be: 1) marginalized; and 2) multiplied

Factor Multiplication

In our illustrative example, we have multiplied two factors:



$P(A)$	
a_1	a_2
0.7	0.3

	$P(B\|A)$	
	b_1	b_2
a_1	0.6	0.4
a_2	0.3	0.7

	$P(a)P(b\|a)$	
	b_1	b_2
a_1	0.6 x 0.7	0.4 x 0.7
a_2	0.3 x 0.3	0.7 x 0.3

Factor Multiplication

Multiplying factors is another factor.

Factor Multiplication

- Definition:
  - Result of multiplying factors $\phi_1(\mathbf{X})$ and $\phi_2(\mathbf{Y})$ is another factor over variables $\mathbf{Z} = \mathbf{X} \cup \mathbf{Y}$:

$$(\phi_1\phi_2)(\mathbf{z}) \equiv \phi_1(\mathbf{x})\phi_2(\mathbf{y})$$

    where **x** and **y** are compatible with **z.**

- Properties:
  1. $\mathrm{dom}(\phi_1\phi_2) = \mathrm{dom}(\phi_1) \cup \mathrm{dom}(\phi_2)$
  2. Commutative: $\phi_1\phi_2 = \phi_2\phi_1$
  3. Associative: $(\phi_1\phi_2)\phi_3 = \phi_1(\phi_2\phi_3)$
  4. Unit factor: **1** contains only 1's and is defined over any domain such that $\mathbf{1}\phi = \phi$ for all factors ϕ.

Factor Marginalization

$$A \rightarrow B \rightarrow C \rightarrow D$$

$P(A)$	
a_1	a_2
0.7	0.3

	$P(B\|A)$	
	b_1	b_2
a_1	0.6	0.4
a_2	0.3	0.7

	$P(a)P(b\|a)$	
	b_1	b_2
a_1	0.6 x 0.7	0.4 x 0.7
a_2	0.3 x 0.3	0.7 x 0.3

Step 1

$$\phi_1(b) = \sum_a P(a)P(b|a)$$

Sum over a

$\phi_1(b)$	
b_1	b_2
0.51	0.49

Factor Marginalization

Factor Marginalization

- Definition:
  - Let ϕ be a factor over variables **X** and X $\in$ **X**.
  - Result of summing out variable X from factor ϕ is another factor ϕ' over variables **Y** = **X** \ {X} :

$$\phi'(\mathbf{Y}) = (\sum_X \phi_X)(\mathbf{y}) = \sum_x \phi(x, \mathbf{y})$$

  - We also say ϕ is projected down to **Y** or that X is marginalized out.

- Properties:
  1. Unit factor property: $\sum_X \phi_X = \sum_x P(X \mid \cdot) = \mathbf{1}$
  2. Distributive: If ϕ_1 and ϕ_2 are factors and if X appears only in ϕ_2:

$$\sum_X \phi_1 \phi_2 = \phi_1 \sum_X \phi_2$$

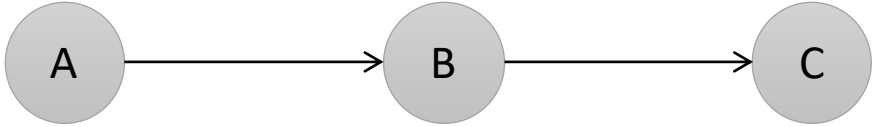
  3. Commutative:

$$\sum_X \sum_Y \phi = \sum_Y \sum_X \phi$$

An Example

Φ(A)=P(A)	
0.6	0.4

	Φ(B,C)=P(C|B)	
b1	0.30	0.70
b2	0.50	0.50



	Φ(A,B)=P(B|A)	
a1	0.90	0.10
a2	0.20	0.80

To compute P(C) by summing over A and then B

Multiply Φ(A) and Φ(A,B):

	Φ(A) Φ(A,B)	
a1	0.54	0.06
a2	0.08	0.32

Summing over A to get Φ(B):

Φ(B)	
0.62	0.38

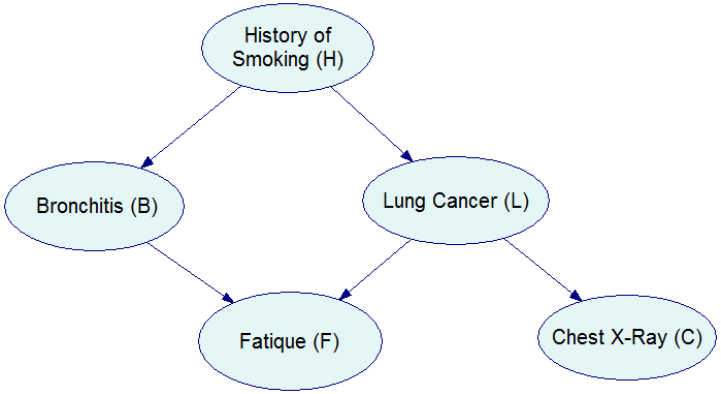
Multiply Φ(B) and Φ(B,C):

	Φ(B) Φ(B,C)	
b1	0.186	0.434
b2	0.190	0.190

Summing over B to get Φ(C):

Φ(C)	
0.376	0.624

Example



$$P(b|h,c) = \alpha P(b,h,c) = \alpha \sum_l \sum_f P(b,l,f,h,c)$$

$$= \alpha \sum_l \sum_f P(h)P(b|h)P(l|h)P(c|l)P(f|b,l)$$

$$= \alpha P(h)P(b|h) \sum_l P(l|h)P(c|l) \sum_f P(f|b,l)$$

$$= \alpha \phi_H(h)\phi_B(b,h) \sum_l \phi(l,h)\phi_C(c,l) \sum_f \phi_F(f,b,l)$$

$$= \alpha \phi_H(h)\phi_B(b,h) \sum_l \phi_L(l,h)\phi_C(c,l)\phi_1(b,l) \qquad \text{(where } \phi_1(b,l) = \textstyle\sum \phi_F(f,b,l)\text{)}$$

$$= \alpha \phi_H(h)\phi_B(b,h)\phi_2(b,h,c) \qquad \text{(where } \phi_2(b,h,c) = \textstyle\sum_l \phi_L(l,h)\phi_C(c,l)\phi_1(b,l)\text{)}$$

Complexity Analysis

- Variable elimination vs. enumeration: $O(n2^k)$ vs. $O(2^n)$

  where: n is no. of nodes in network

  k is max. no. of variables in a factor

- Time and space requirements dominated by:
  - size of interim factors constructed
  - elimination order of variables
- Amount of work bounded by the width – size of largest factor created in elimination order

Causal Reasoning

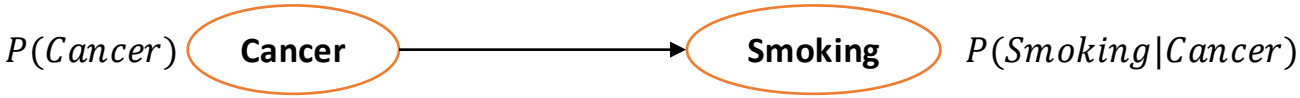
Causal networks

Do-Calculus

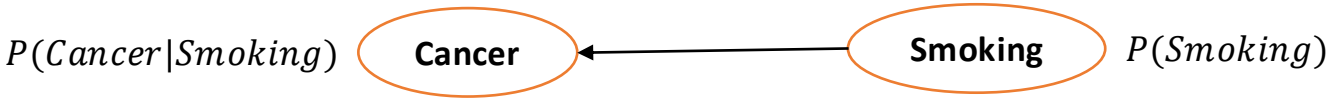
Causal Inference

- Estimating treatment/causal effect of an intervention on an outcome.
  - Examples:
    - Effect of smoking on the risk of cancer.
    - Effect of different advertising campaigns on the revenue of a company.
    - Effect of a business solution on the revenue of a client.
    - Effect of a new medicine/vaccine on patient's health.
- Causal inference can help reduce bias in AI models.

Causal Ordering of Variables



$$P(Smoking, Cancer) = P(Smoking|Cancer)P(Cancer)$$

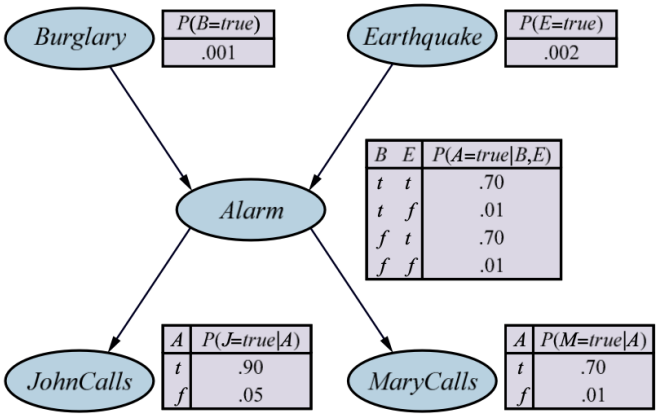


$$P(Smoking, Cancer) = P(Cancer|Smoking)P(Smoking)$$

- These two Bayesian networks are equivalent, hence convey the same information.
- How could they convey the same information when we know that *Smoking* causes *Cancer* and not the other way around?
- These two equivalent Bayesian networks do not encode causal information.
- Hence, we cannot perform causal analysis with Bayesian Networks.

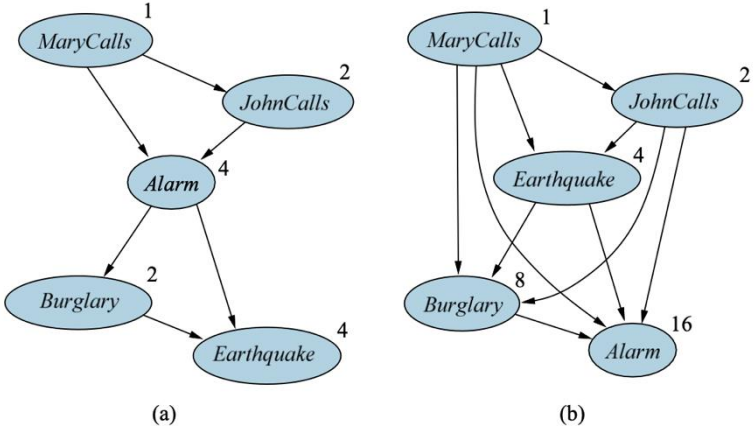
Causal Ordering of Variables

How many parameters are required?



Source: RN Figure 13.2

How many parameters are required?



Source: RN Figure 13.3

Compact network structures

Causal Ordering of Variables

- Advantages of keeping node ordering in Bayes nets compatible with the direction of causation:
  - Easier to assess conditional probabilities
  - Compact network structures
  - Causal analysis

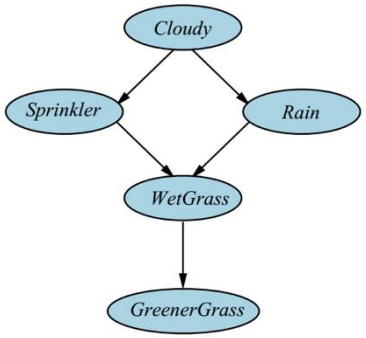
Causal Networks

- Causal Bayesian networks:
  - Restricted class of Bayesian networks that forbids all but causally compatible orderings.
  - To decide on arrow directionality by considerations beyond probabilistic dependence and invoke a different type of judgment
- Example:

  Fire → Smoke? or Smoke → Fire?

Structural Equation

- Structural equations describe causal relationship among variables in a causal network. It has the following form:
  - $X_i = f_i(OtherVariables)$

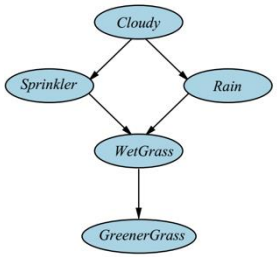


$$
\begin{aligned}
C &= f_C(U_C) \\
R &= f_R(C, U_R) \\
S &= f_S(C, U_S) \\
W &= f_W(R, S, U_W) \\
G &= f_G(W, U_G)
\end{aligned}
$$

- The *U*-variables represent unmodeled variables (also called error terms or disturbances) that perturb the functional relationship between each variable and its parents.
- System of structural equations allows us to predict how interventions will affect the operation of the system and hence the observable consequences of those interventions. This is not possible given just the joint distribution.

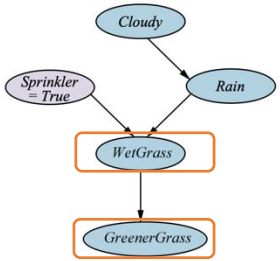
Intervention/do-Operator

- Intervention:
  - Turn the sprinkler on: do(Sprinkler = True)



$$
\begin{aligned}
C &= f_C(U_C) \\
R &= f_R(C, U_R) \\
S &= f_S(C, U_S) \\
W &= f_W(R, S, U_W) \\
G &= f_G(W, U_G)
\end{aligned}
$$

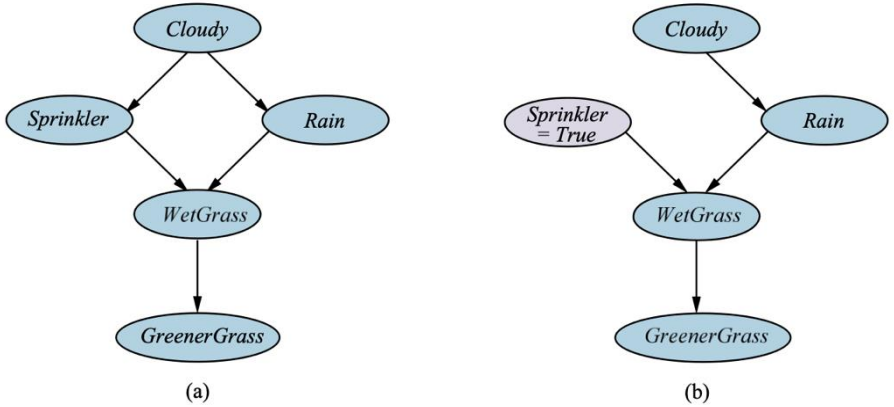
$$
\begin{aligned}
C &= f_C(U_C) \\
R &= f_R(C, U_R) \\
S &= true \\
W &= f_W(R, S, U_W) \\
G &= f_G(W, U_G).
\end{aligned}
$$



- New joint distribution:
  - For remaining variables conditioned on do(Sprinkler = True)

$$P(c, r, w, g \mid do(S = true) = P(c)\, P(r \mid c)\, \boxed{P(w \mid r, s = true)}\, \boxed{P(g \mid w)}$$

Action and Observation



- Conditional distribution ≠ Interventional distribution
- For example:

$$P(C = true | S = true) \neq P(C = true | do(S = true))$$

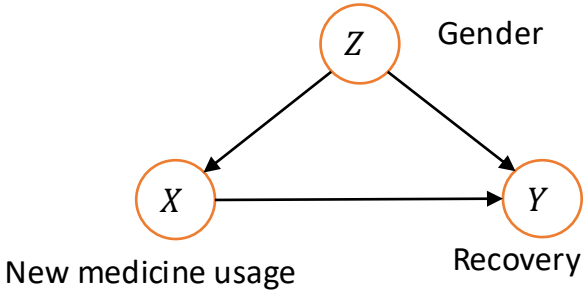
  - Suppose that:

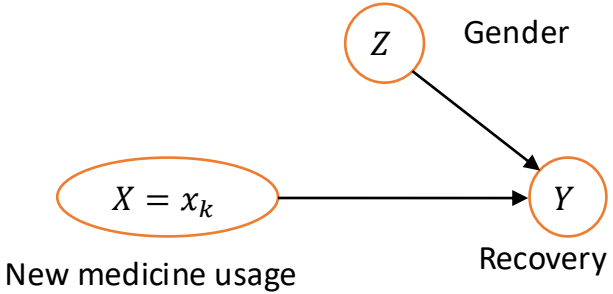
$$1 > P(C = true) > 0,$$

$$S = f_S(C, U_S) = \neg C$$ (The sprinkler is turned on when it is not cloudy)

  - Then, $P(C = true | S = true) = 0$ (In Figure (a))
  - However, $P(C = true | do(S = true)) = P(C = true)$ (In Figure (b))

Effects of the do-Operator





$$P(x,y,z) = P(z)P(x|z)P(y|z,x)$$

$$= \begin{cases} 1 & if\ x = x_k \\ 0 & if\ x \neq x_k \end{cases}$$

$$P_{x_k}(x,y,z) = P_{x_k}(z)P_{x_k}(x)P_{x_k}(y|z,x) \quad = P(z)P_{x_k}(x)P(y|z,x_k)$$

$$= \begin{cases} P(z)P(y|z,x) = \dfrac{P(x,y,z)}{P(x|z)} & \text{if } x = x_k \\ 0 & \text{if } x \neq x_k \end{cases}$$

We generalize this formula in the next slide.

Effects of the do-Operator (cont.)

- In a general causal network with variables $X_1, \ldots, X_n$.
  - The network corresponds to a joint distribution:

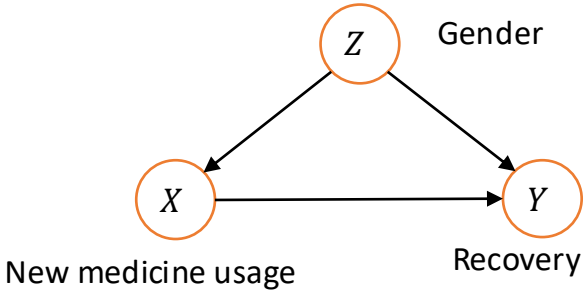
$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \mid parents(X_i)).$$

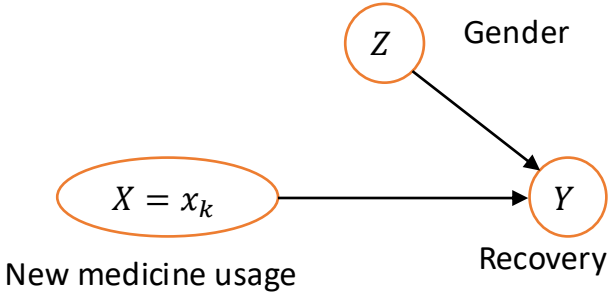
  - After applying $do(X_j = x_{jk})$, the new joint distribution $P_{x_{jk}}$ simply omits the factor for X_j:

$$P_{x_{jk}}(x_1, \ldots, x_n) = \begin{cases} \prod_{i \neq j} P(x_i \mid parents(X_i)) = \frac{P(x_1, \ldots, x_n)}{P(x_j \mid parents(X_j))} & \text{if } x_j = x_{jk} \\ 0 & \text{if } x_j \neq x_{jk} \end{cases}$$

  - This follows from the fact that setting X_j to a particular value x_j corresponds to deleting the equation $X_j = f_j(parents(X_j), U_j)$ from the system of structural equations and replacing it with $X_j = x_{jk}$.

Effects of the do-Operator (cont.)





$$P(x, y, z) = P(z)P(x|z)P(y|z, x)$$

$$P(y, z|do(X = x_k)) \stackrel{\text{def}}{=} P_{x_k}(x_k, y, z) = P(z)P(y|z, x_k)$$

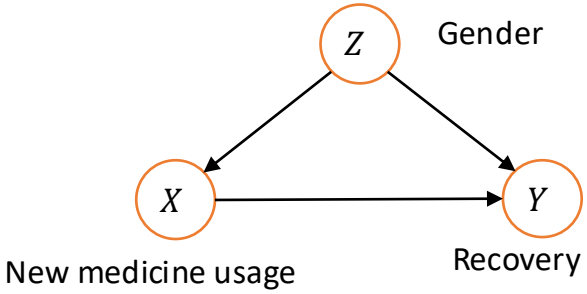
Marginalize out z:

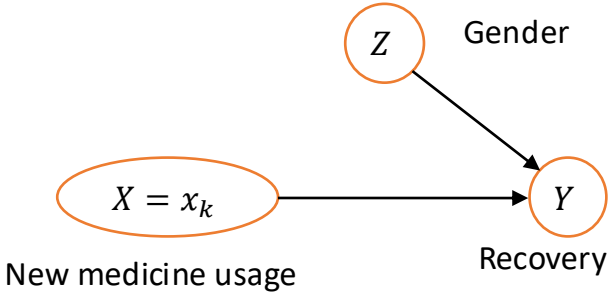
$$P(y|do(X = x_k)) = \sum_z P(z)P(y|z, x_k)$$

This is the adjustment formula to estimate causal effects from observational data.
Average treatment effect (ATE) of using the new medicine:

$$\text{ATE} = E[y|do(X = true)] - E[y|do(X = false)]$$

The Backdoor Criterion





The backdoor adjustment formula:

$$P(y|do(X = x_k)) = \sum_z P(z)P(y|z, x_k)$$

Definition: A set of variables Z satisfies the back-door criterion relative to an ordered pair of variables (X, Y) in a DAG G if:
(i) no node in Z is a descendant of X; and
(ii) Z blocks every path between X and Y that contains an arrow into X.

The Backdoor Criterion

- Advantage of the backdoor criterion
  - A golden method to estimate causal effects is the randomized control trial (RCT, also called randomized experiment), e.g., estimate effectiveness of a specific vaccine such as Covid-19 vaccine, estimate causal effects of a medicine, etc.
  - However, some real-life scenarios, RCT is impossible. For example, RCT is impossible to estimate effect of smoking on cancer (randomly assign participants to smoking/non-smoking is unethical), RCT is also impossible to estimate causal effects of air quality on asthma attack (we cannot randomly assign different level for air quality). In these cases, the backdoor criterion might be helpful if we can find the variables satisfying backdoor criterion.
  - If we can find the set of variables satisfying backdoor criterion **and they are observed** , then the causal effects can be estimated from observational data using the adjustment formula.
- Disadvantage of the backdoor criterion
  - If there are some unobserved variables in the above set, then it is impossible to use backdoor adjustment.
  - In this case, one can consider the front-door adjustment formula.

Causal Inference Applications in the Industry

- Conditional Average Treatment Effects
  - Leveraging Causal Modeling to Get More Value from Flat Experiment Results - **Doordash (2020)**
  - Free Lunch! Retrospective Uplift Modeling for Dynamic Promotions Recommendation within ROI Constraints - **Booking (2020)**
- Interference
  - How Meta Tests Products with Strong Network Effects - **Meta (2023)**
  - Experimental Design in Two-Sided Platforms: An Analysis of Bias - **Airbnb (2021)**
  - Design and analysis of experiments in networks: Reducing bias from interference - **Facebook (2014)**

Causal Inference Applications in the Industry

- Pricing
  - Causal Forecasting for Pricing - **Zalando (2024)**
  - Science of Price Experimentation at Amazon - **Amazon (2024)**
- Experimentation Platform
  - On the Limits of Regression Adjustment - **Meta (2023)**
  - Decision Making at Netflix - **Netflix (2021)**
  - Spotify's New Experimentation Platform **- Spotify (2020)**
  - Under the Hood of Uber's Experimentation Platform - **Uber (2018)**
  - Democratizing Online Controlled Experiments at Booking.com - **Booking (2017)**

Causal Inference Applications in the Industry

- Trustworthy Experiments
  - Answering Counterfactual “what-if” Questions in a Trustworthy and Efficient Manner - **Spotify (2023)**
  - Imbalance Detection for Healthier Experimentation - **Etsy (2022)**
  - Patterns of Trustworthy Experimentation: During-Experiment Stage - **Microsoft (2021)**
  - More Trustworthy A/B Analysis: Less Data Sampling and More Data Reducing - **Microsoft (2021)**
  - Data Quality: Fundamental Building Blocks for Trustworthy A/B testing Analysis **- Microsoft (2021)**

Summary

- Exact Inference in Bayesian Networks
  - Inference by enumeration
  - Factor or potential algebra
  - Variable elimination
- Causal reasoning in Bayesian Networks
  - Causal ordering of variables
  - Causal networks or causal diagrams
  - Do-calculus
  - The Backdoor criterion

Homework

- Readings:

1. RN: 13.3, 13.5

- Exploration:
  - Build a small Bayesian network on GINIE or SMILE
  - Run the different inference (solution) algorithms on a few examples
  - Front-door adjustment formula

References

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  - (RN) Russell, S. and P. Norvig, Artificial intelligence: A modern approach. 4th ed. (Global ed.) 2021: Pearson. [Kindle Edition] (Alternate: 3rd ed.)
    [Table of contents for 4th ed.: http://aima.cs.berkeley.edu/contents.html]
  - (M) Murphy, K.P., Probabilistic Machine Learning: Advanced Topics. 2023: MIT Press. [Free copy for personal use: https://probml.github.io/pml-book/book2.html]
- Other references:
  - (H) Heckerman, D., A Tutorial on Learning With Bayesian Networks. 2020. https://arxiv.org/abs/2002.00269
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  - https://doi.org/10.1017/CBO9780511811357
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